

Multiple Linear Regression Analysis: Computer Hardware Performance

IEE 380: Probability and Statistics for Engineering Problem Solving

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1. Introduction and Data Description

Dataset Description

For this project, I selected the Computer Hardware Performance dataset from the options provided. I obtained the data from Kaggle. It has 209 observations, meaning there are 209 different computer systems from 30 different hardware vendors like IBM, Microdata, and Honeywell. The objective of this project is to use multiple linear regression to determine the Published Relative Performance (PRP) of each computer system based on its hardware specifications.

Variables

Variable	Description	Unit
MYCT	Machine cycle time	Nanoseconds
MMIN	Minimum main memory	Kilobytes
MMAX	Maximum main memory	Kilobytes
CACH	Cache memory size	Kilobytes
CHMIN	Minimum channels	Count
CHMAX	Maximum channels	Count
PRP	Published Relative Performance	No unit
ERP	Estimated Relative Performance	No unit

The table above explains our variables in the dataset as well as the units they're measured in. The dataset has a total of 10 columns. For this project, I only used 7 of them. PRP is our response variable, which is what we are trying to predict. I didn't use ERP in this project because it's an estimate of PRP, meaning it's not an actual hardware specification. I also didn't use the Vendor or Model column because those contain text characters which can't be used in regression.

Sampling Method

The software of my choice was Minitab Statistical Software. I used this software to take a random sample of 100 observations which was taken from the full dataset of 209. To do this, I imported my data and selected Calc > Random Data > Sample From Columns > set the number of rows to sample to 100. I then

selected all the columns and stored the results in the new columns. The sample was taken without replacement.

2. Exploratory Data Analysis

Summary Statistics

To get started with the exploratory analysis, I first went over the summary statistics for all 7 variables so I could get a better understanding of the data before starting the regression.

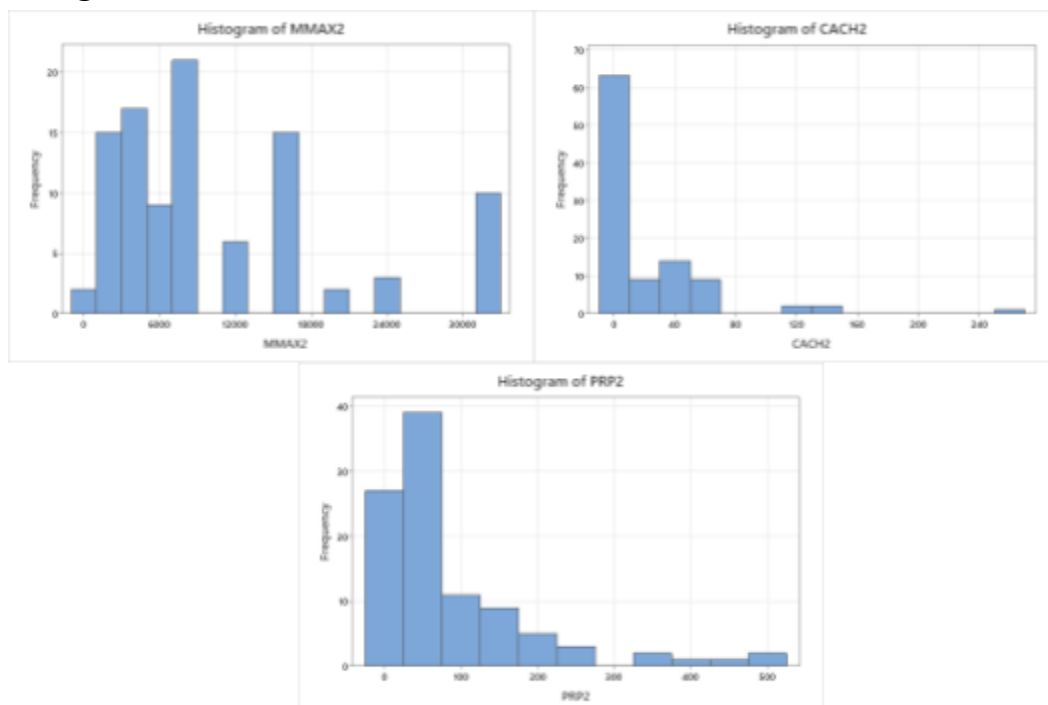
Descriptive Statistics: MYCT2, MMIN2, MMAX2, CACH2, CHMIN2, CHMAX2, PRP2

Statistics

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
MYCT2	100	0	206.71	27.1773	271.773	17	50	105	223.75	1500
MMIN2	100	0	2716.32	323.472	3234.72	64	585	2000	4000	16000
MMAX2	100	0	10546.4	918.891	9188.91	64	4000	8000	16000	32000
CACH2	100	0	21.55	3.80376	38.0376	0	0	8	32	256
CHMIN2	100	0	4.27	0.644927	6.44927	0	1	1	6	52
CHMAX2	100	0	13.32	1.57966	15.7966	0	4	8	16	104
PRP2	100	0	88.37	10.7410	107.410	6	24	45	112.5	510

One thing that caught my eye was the PRP2. It has a mean of 88.37 but only a median of 45. This tells me that most of the computer systems in our sample are performing on an average level, while there are a few high performing systems. These high performing systems help in pulling the average up. We can also see this in our maximum PRP2 value, which is 510. We also see this with our MMAX2 since it has a minimum of 64 KB and a maximum of 32000 KB which tells me the sample has both low and high end computer systems. With our CACH2, we can see that the minimum is 0, but we have a median of 8, this tells me that most systems in the sample have very little cache memory.

Histograms



When looking at the histograms above, all three variables display a right skewed distribution. Most of the values are on the lower end, with only a few on the higher end. For PRP2, most systems have a performance score between 0-100, with only a few systems reaching a score between 400-500. For MMAX2, we can see that most systems have lower memory ranging from 0 KB to 8000 KB while there are a few systems that range up to 32,000 KB. Out of all the histograms, CACH2 is the most skewed. This is because most machines have little to no cache memory.

The histograms for all 7 variables can be found in the Appendix.

Correlation Analysis

To complete the exploratory analysis, I ran a correlation analysis in Minitab. This allowed me to see how each variable relates to PRP before starting the regression.

Correlations

	MYCT2	MMIN2	MMAX2	CACH2	CHMIN2	CHMAX2
MMIN2	-0.367					
MMAX2	-0.417	0.755				
CACH2	-0.295	0.665	0.574			
CHMIN2	-0.289	0.436	0.602	0.575		
CHMAX2	-0.323	0.328	0.516	0.500	0.757	
PRP2	-0.363	0.853	0.824	0.809	0.682	0.539

When looking at the image above, MMIN2 has the strongest correlation out of all the 7 variables with PRP2. Followed by MMAX2 at 0.824 and CACH2 at 0.809. This tells me that when it comes to a system's performance, memory and cache have a significant impact. We can also see that MYCT2 has a negative correlation with PRP2 at -0.363. This is because a CPU with a faster cycle time has better performance. If we look at MMIN2 and MMAX2, we can see that they have a high correlation at 0.755. I believe this is a sign of multicollinearity, which I checked further in the project using VIF.

3. Model Development and Refinement

Initial Model

To begin the model, I initially ran a regression test in Minitab using all six predictor variables with PRP2 as the response variable. This allowed me to see which variables were significant and which were not.

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	6	1035734	172622	150.84	0.000
MYCT2	1	727	727	0.64	0.428
MMIN2	1	53015	53015	46.33	0.000
MMAX2	1	24194	24194	21.14	0.000
CACH2	1	47618	47618	41.61	0.000
CHMIN2	1	17338	17338	15.15	0.000
CHMAX2	1	223	223	0.20	0.660
Error	93	106427	1144		
Lack-of-Fit	88	106350	1209	78.31	0.000
Pure Error	5	77	15		
Total	99	1142161			

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	-11.65	7.29	-1.60	0.113	
MYCT2	0.0112	0.0140	0.80	0.428	1.25
MMIN2	0.01265	0.00186	6.81	0.000	3.13
MMAX2	0.003013	0.000655	4.60	0.000	3.14
CACH2	0.865	0.134	6.45	0.000	2.25
CHMIN2	3.495	0.898	3.89	0.000	2.90
CHMAX2	-0.151	0.342	-0.44	0.660	2.52

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
33.8287	90.68%	90.08%	86.67%

Looking at the results of the model summary, we can see that R-squared has a score of 90.68%. This shows that the model explains about 90.68% of the variance in PRP2. If we look at the analysis of variance, specifically the regression row, the overall model had an F-Value of 150.84 and a P-Value of 0.000. This means that our model is statistically significant.

To figure out which variables to remove, I looked at the coefficients table and chose MYCT2 and CHMAX2. This is because MYCT2 has a P-value of 0.428 and CHMAX2 has a P-value of 0.660. Both of these are over the 0.05 limit which means they are not statistically significant. We can also see that with CHMAX2, it has a coefficient value of -0.151 which means it's hurting PRP2 by -0.151. Realistically speaking, the CHMAX2 value made zero sense to me as more channels should increase performance not decrease it, so that's another reason I removed it.

Final Model

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	4	1034641	258660	228.54	0.000
MMIN2	1	55576	55576	49.10	0.000
MMAX2	1	23161	23161	20.46	0.000
CACH2	1	47538	47538	42.00	0.000
CHMIN2	1	22899	22899	20.23	0.000
Error	95	107521	1132		
Lack-of-Fit	77	97226	1263	2.21	0.030
Pure Error	18	10295	572		
Total	99	1142161			

Coefficients

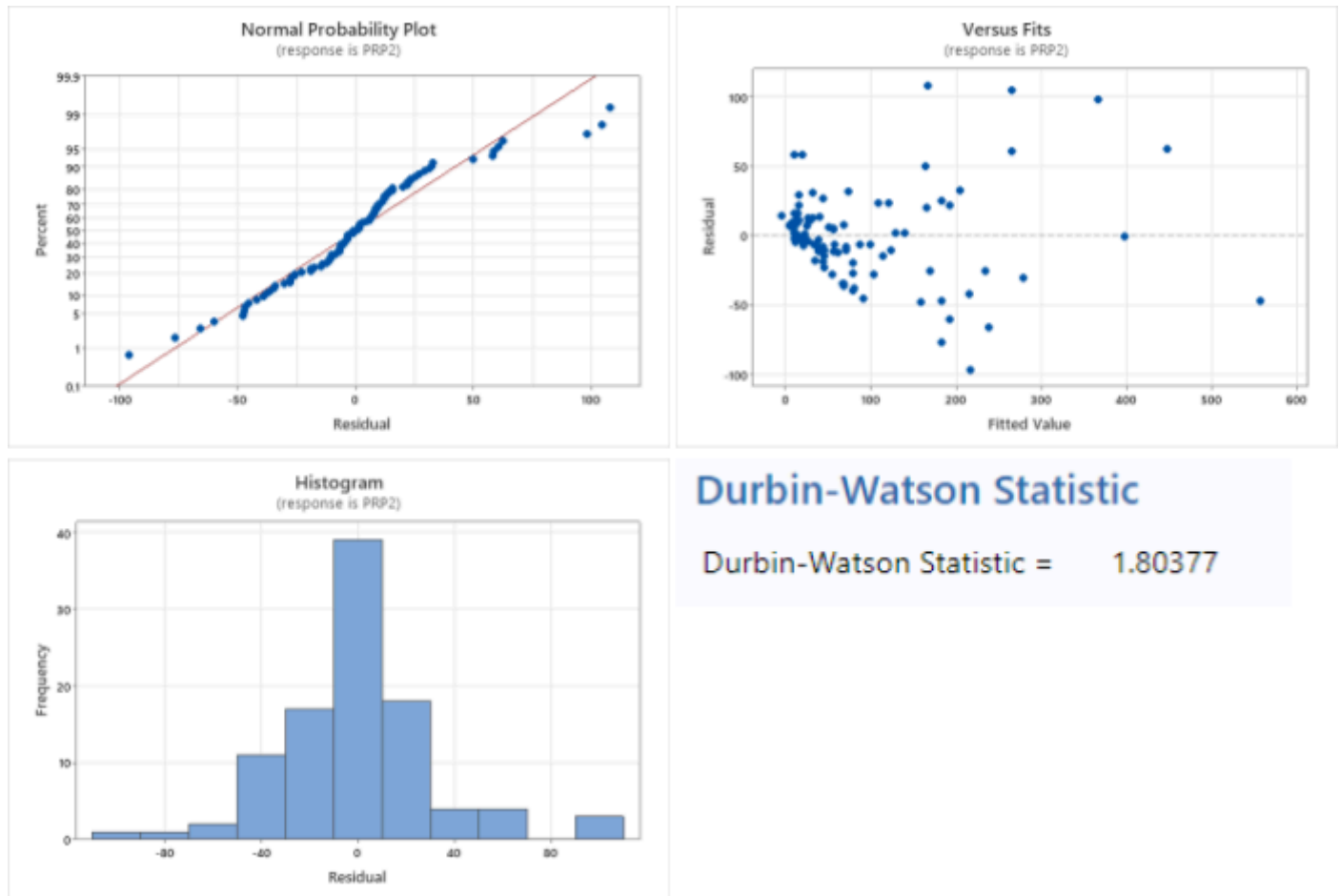
Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	-8.58	5.15	-1.67	0.099	
MMIN2	0.01270	0.00181	7.01	0.000	3.01
MMAX2	0.002868	0.000634	4.52	0.000	2.97
CACH2	0.854	0.132	6.48	0.000	2.20
CHMIN2	3.234	0.719	4.50	0.000	1.88

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
33.6422	90.59%	90.19%	87.60%

Based on the model summary above, we can see that our R-squared has a value of 90.59% with a decrease of 0.09% from our initial model. This tells me that removing MYCT2 and CHMAX2 had very little impact on the model which tells me they were not significant predictors. We can also see that our adjusted R-squared went up by 0.11% from the initial 90.08% which also helps this decision. The analysis of variance shows that the overall model has an F-value of 228.54 and a P-value of 0.000. This tells me that the final model is more significant than the initial. In the P-Value column, we can see that all 4 remaining predictors have a score of 0.000 meaning they are all statistically significant. In the VIF column in the coefficients table, we can see all 4 predictors achieve a score below 5, this means that multicollinearity isn't a concern in the final model.

Adequacy Check



Linearity: In our Residuals vs Fits plot, the residuals are scattered randomly close to zero. This means that our linearity assumption is satisfied.

Independence: The Durbin-Watson statistic has a value of 1.80377 which is close to the value of 2 meaning the independence assumption is satisfied.

Homoscedasticity: Looking at the Residuals vs Fits plot, the residuals start to spread out at a greater fitted value which could suggest a minor issue but nothing to be worried about.

Normality of Residuals: Looking at the Normal Probability Plot, it shows a straight line in which the residuals follow with some deviation at the bottom and top. Our Histogram also appears to be bellshaped, this is caused by a few high performance systems in our sample.

4. Final Model Results and Interpretation

Model Fit

My final model achieved an R-Squared value of 90.59%. This tells us that the model explains 90.59% of the variation in PRP using only 4 hardware variables (MMIN, MMAX, CACH, and CHMIN). The adjusted R-Squared achieved a value of 90.19% which tells us the model is not overfitted. The F-Value achieved a value of 228.54 and the P-Value at 0.000. Since the P-value is under 0.05, our final model is statistically significant.

Analysis of Coefficients

Regression Equation

$$\text{PRP2} = -8.58 + 0.01270 \text{ MMIN2} + 0.002868 \text{ MMAX2} + 0.854 \text{ CACH2} + 3.234 \text{ CHMIN2}$$

This is our regression equation for the final model. The beginning value is our constant value of -8.58 which is the starting point of the equation. This is when all the predictor variables are equal to zero. MMIN holds a value of 0.01270, which means that for every 1 KB increase of minimum memory, the PRP increases by 0.01270. Minimum memory sets the minimum capacity a system can handle, with more minimum memory, the system can handle bigger tasks which improves the performance. MMAX has a value of 0.002868, MMAX sets the maximum capacity a system can handle, 1 KB increase of maximum memory, PRP increases by 0.002868. A system with more maximum memory can handle larger tasks. The CACH has a coefficient of 0.854. CACH is the fastest type of memory in a system, because it is between the CPU and main memory which allows for faster data access. For 1 KB of increase in cache, PRP goes up by 0.854. CHMIN is our minimum channel which holds a value of 3.234, for every new channel, PRP goes up by 3.234. Channels manage the amount of data that transfer between the CPU and other parts of the system. More channels allow for more data to move at once, which improves the overall performance of a system.

5. Discussion and Conclusion

Key Findings

Started the project with six predictor variables, but only used four due to others not being significant. Those 4 variables are MMIN, MMAX, CACH, and CHMIN. Out of all the variables, CACH has the most impact on PRP because cache assists the CPU in accessing data faster. The final model achieved an R-Squared value of 90.59% which I would say is strong.

Limitations

I noticed a few limitations in this project. Some of the residuals deviated from the line in the Normal Probability Plot. This is because of the few high performance systems in the sample. There were also some minor issues with the Homoscedasticity in the Residuals vs Fits plot.

Conclusion

The final model explains 90.59% of the variation in PRP while only using four variables. If you have a system and you know the machine's minimum memory, maximum memory, cache size, and number of channels you are able to predict its performance with this model.

Appendix

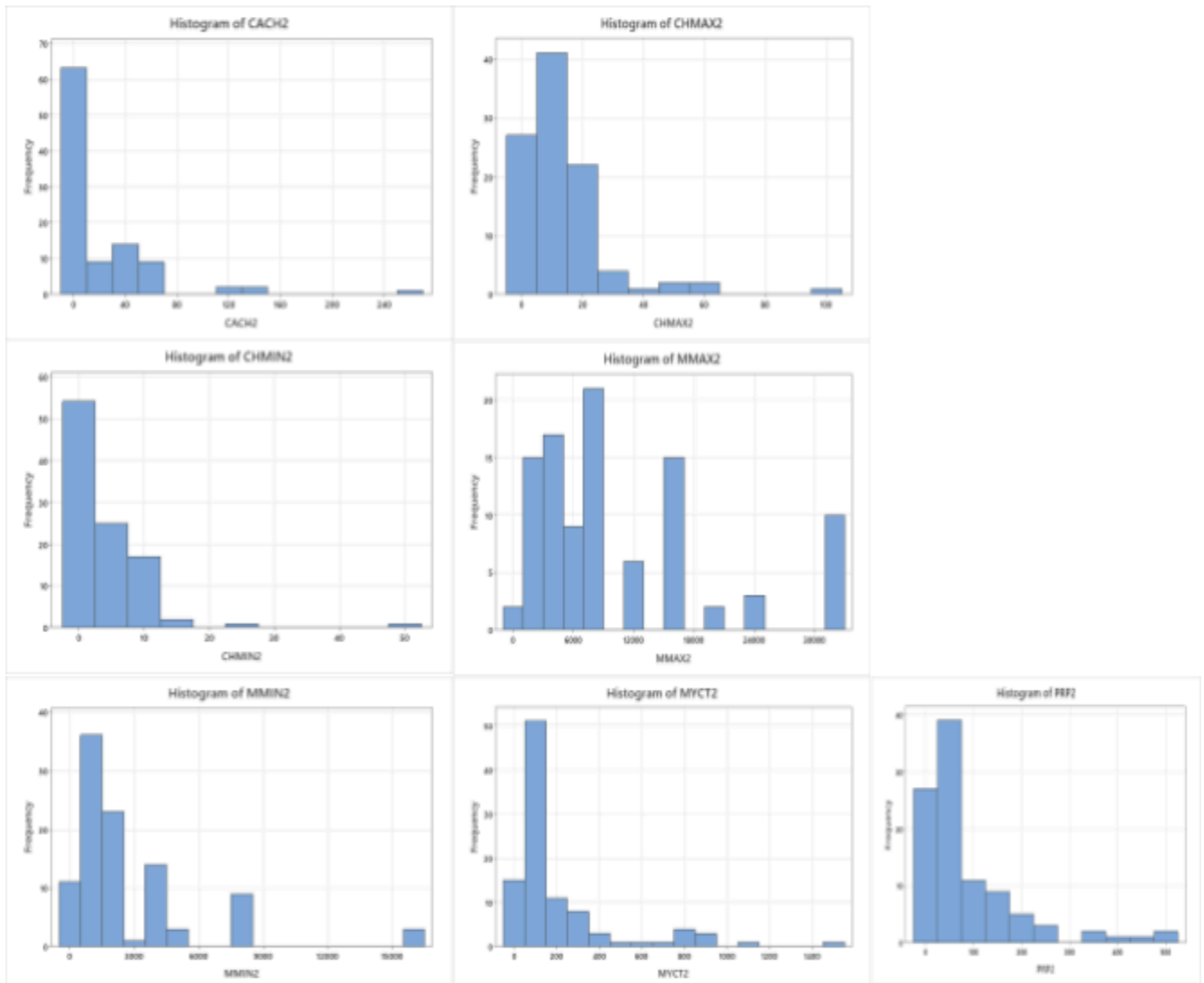
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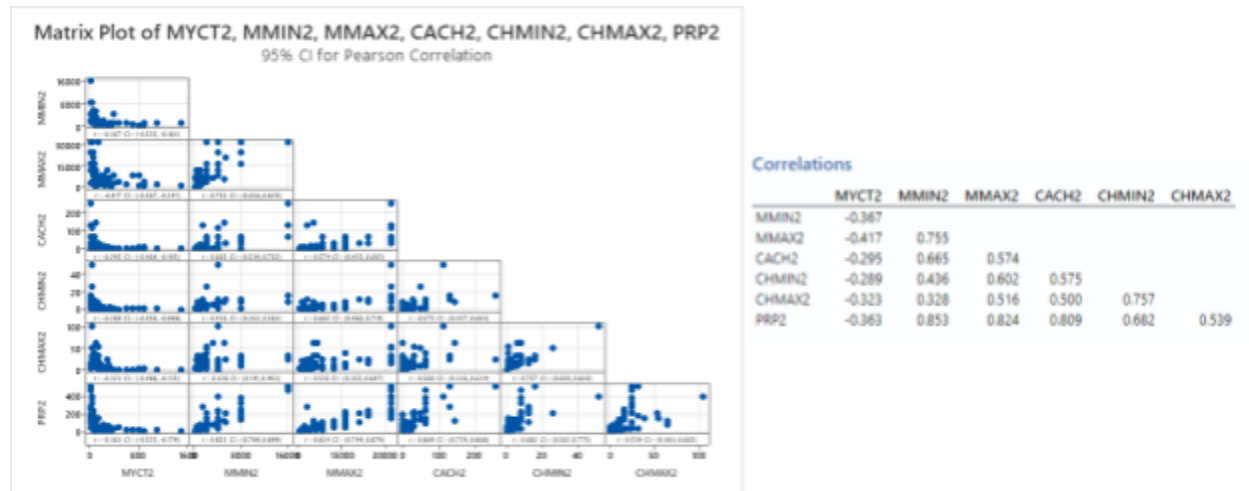
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PRP2	100	0	88.37	10.7410	107.410	6	24	45	112.5	510

Histogram of all 7 variables



Correlation Table and Matrix Plot



Initial Regression Output (6 Predictors)

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Model Summary

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Final Regression Output (4 Predictors)

Analysis of Variance

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Regression	4	1034641	258660	228.54	0.000
MMIN2	1	55576	55576	49.10	0.000
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Diagnostic Plots

